

From Predictive Importance to Decision Value: Evaluating XAI-Based Weights for ITS Prioritisation

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Abstract. Feature attributions from predictive models are increasingly rescaled into global criterion weights for multi-criteria decision-making. Predictive importance and decision value are, however, different quantities, and a benchmark must be able to tell them apart before it can compare methods. We build a controlled synthetic laboratory for Intelligent Transportation Systems prioritisation: six alternatives, ten criteria, a nonlinear oracle, XGBoost, interventional TreeSHAP, six structured global-weight pipelines and benefit-oriented MOORA. A structural audit first showed that the original generator’s apparent contextual variation cancelled exactly under the normalization used by the decision operator. A corrected monotone response kernel removed that algebraic collapse. We then executed a five-world adequacy gate that had been frozen before execution. Four structured methods, including SHAP, cleared the prospectively specified RandomWeights adequacy criterion; SHAP reached a median Kendall τ_b improvement of 0.288, positive in all five worlds. No structured method cleared the MajorityWinner criterion—SHAP’s median Top-1 and regret advantages against that constant reference were exactly zero—and closed-form oracle attribution failed the same gate, so the outcome is not an artefact of learned approximation. The gate therefore stopped the protected primary factorial. Recovering structured importance is not sufficient: synthetic XAI-MCDM benchmarks should demonstrate incremental decision value against meaningful simple references before comparing methods.

Keywords: Explainable AI · Multi-Criteria Decision-Making · Benchmark Validity · Prospective Stopping · Decision Support · Intelligent Transportation Systems

1 Introduction

Transport agencies routinely prioritise Intelligent Transportation Systems (ITS) with limited operational evidence and limited capacity for formal preference elicitation. The task is naturally multi-criteria: candidate deployments differ in mobility, reliability, person-throughput, safety, environmental performance, accessibility, readiness, interoperability and lifecycle burden. Machine learning and explainable artificial intelligence (XAI) offer an attractive shortcut. Fit a performance model, aggregate its feature attributions into criterion “importance”,

normalize the result into weights, and rank alternatives with a multi-criteria decision-making (MCDM) rule.

That shortcut contains a semantic leap and an empirical one. MCDM weights take their meaning from criterion scales and from the aggregation model; they are not generic measures of importance [1,2,3]. SHAP values allocate model output under a specified cooperative-game functional [6], but a predictive attribution is not automatically a stakeholder preference, a causal effect or a valid trade-off constant [8,9,10]. Consequently

predictive attribution $\longrightarrow$ global criterion weight $\longrightarrow$ MCDM ranking

is a pipeline to be validated, not an identity to be assumed.

Controlled synthetic studies look well suited to that validation, because the data-generating process and an oracle reference are both known. Yet a synthetic benchmark can carry a great deal of raw variation and still pose a nearly trivial decision problem, and a method comparison run on such a benchmark measures implementation differences rather than decision value. Benchmark studies are also vulnerable to over-optimism when generators, estimands or analyses are revised after method performance has been observed [23,24].

We therefore separate two notions and use them consistently throughout. *Validity* refers to the conceptual scientific properties of the benchmark: whether its responses vary, whether that variation survives normalization, whether the oracle is well defined. *Adequacy* refers to the operational, prospectively frozen test used to decide whether a protected primary comparison is warranted at all. A benchmark can be valid and still be inadequate.

We address two prospective research questions.

- RQ1.** Can structural diagnostics reveal when apparent contextual variation disappears under the normalization used by the downstream MCDM operator?
- RQ2.** After correcting that structural defect, do structured global-weight pipelines provide sufficient incremental decision value over random-weight and constant-decision references to justify protected primary evaluation?

After the study was permanently closed we additionally froze one descriptive diagnostic, reported here under a deliberately separate label.

- DQ1 (post-closure, descriptive).** After Pilot B was permanently closed, what do the archived context-level selection maps reveal about the observed decision behaviour of the seven frozen fixed-weight vectors?

DQ1 is post-closure and descriptive. It is non-confirmatory, and it cannot modify, reinterpret or reopen the Pilot-B gate. It is reported because it is explanatory, not because it carries evidential weight.

The paper contributes a validation workflow rather than a new weighting algorithm. It separates raw variation, structural non-separability and downstream decision discrimination and shows that the first two do not imply the third; it carries a closed-form oracle attribution alongside learned SHAP, separating estimation error from losses created by global compression and aggregation; and it

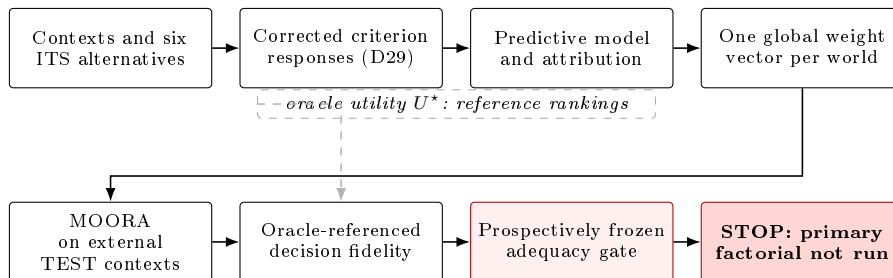


Fig. 1. Study logic. Estimation partitions (FIT and WEIGHT) support model fitting and global-weight estimation; disjoint external TEST contexts support MOORA decisions and oracle-referenced fidelity. The gate decides whether the protected primary comparison is scientifically warranted; here it decided against.

demonstrates a prospectively frozen adequacy gate that stops a planned comparison when its decision premise fails. To our knowledge this is a rare prospectively gated controlled evaluation of whether global attribution-derived weights retain downstream decision value after MCDM aggregation. The generator genealogy, candidate grids, seed namespaces, software tests, frozen protocols and unexecuted analyses are preserved in the companion supplement.

Figure 1 gives the study logic end to end. Three features are central: estimation data and external TEST contexts are disjoint, an oracle reference runs in parallel with the learned pipeline, and the final gate is evaluated before the protected experiment is allowed to start.

2 Related Work and Contribution Boundary

XAI-MCDM studies have shown that model explanations can be integrated with ranking methods across domains including transport and infrastructure [14,15,16], while objective weighting methods such as CRITIC and entropy derive weights from dispersion and conflict within a decision matrix [17,18,4]. These families answer different questions: attribution summarizes a predictive functional, regression and permutation summarize supervised association, and objective weights summarize the observed criterion matrix. Their outputs are numerically comparable but are not interchangeable notions of preference.

Synthetic XAI benchmarks typically compare explanations against known attribution targets [12,13]. Our focus lies one step downstream. Even exact recovery of a global attribution functional need not recover context-specific oracle choices once that functional has been compressed into a single fixed weight vector and passed through an aggregation rule. Prediction, attribution and decision fidelity can fail at different stages, so we evaluate learned SHAP alongside a matched closed-form oracle attribution using ranking, Top-1 and regret metrics.

The contribution boundary is deliberately narrow. This study does not estimate empirical ITS effects, does not elicit stakeholder preferences, does not

Table 1. ITS alternatives in the controlled decision laboratory.

ID	Alternative	Operational scope
A_1	ATSC	Adaptive phase, split, offset or cycle decisions.
A_2	TSP	Transit-priority requests and signal responses.
A_3	TIDM	Incident detection, verification, response and recovery.
A_4	RMTI-DRG	Real-time multimodal information and dynamic route guidance.
A_5	V2X-CS	Cooperative safety warnings, excluding signal timing and TSP.
A_6	DSM	Dynamic speed management and harmonisation.

identify a preferred real-world intervention, and does not test whether SHAP works in general. ITS supplies a controlled applied decision setting. The scientific object is the adequacy of a global-weight XAI-MCDM benchmark at one prospectively specified development reference condition.

3 Benchmark and Validation Design

3.1 An ITS Decision Laboratory

The benchmark compares six deployment packages at a common first-stage prioritisation level (Table 1). They represent deployable ITS and transport-systems-management families rather than city-specific calibrated effects [20,21,22].

Ten benefit-oriented criteria cover mobility efficiency, reliability, person-throughput, safety, resilience, environmental efficiency, accessibility, readiness, interoperability/scalability and direction-adjusted lifecycle burden, driven by five correlated latent factors: demand, incidents, person movement, environmental sensitivity and infrastructure readiness.

Each replication seed defines one benchmark world: its technology parameters, its contexts and all of its random streams. The world is the statistical unit throughout this paper. Alternative-context rows are never treated as independent population replications.

3.2 A Structural Defect and the D29 Correction

The initial noise-free response for the seven active criteria C_1 – C_7 had the form

$$g_{asj} = \theta_{aj} o_{sj}, \quad (1)$$

where a , s and j index alternative, context and criterion, θ_{aj} is a capability amplitude and o_{sj} a context opportunity. Under the vector normalization used by the decision operator, for $o_{sj} > 0$,

$$r_{asj} = \frac{\theta_{aj} o_{sj}}{\sqrt{\sum_b \theta_{bj}^2 o_{sj}^2}} = \frac{\theta_{aj}}{\sqrt{\sum_b \theta_{bj}^2}}. \quad (2)$$

The common opportunity term cancels exactly, so every context yields an identical normalized column. Sum, max and min–max normalizations exhibit the same common-scale cancellation. This answers RQ1: raw contextual variation, on its own, was not evidence that the benchmark posed a context-sensitive decision problem, and only a diagnostic applied *at the operator’s normalization* could reveal it.

After prospectively separated calibration and eligibility screening, the frozen correction—referred to as the D29 kernel—for the active C_1 – C_7 pathways was

$$g_{asj} = \theta_{aj} o_{sj} + 0.05 v_{aj} o_{sj} (1 - o_{sj}), \quad v_{aj} \in [-1, 1]. \quad (3)$$

The deterministic contrast values v_{aj} were assigned without any outcome or winner information. Because active $\theta_{aj} \geq 0.05$, the response stays monotone on $o \in [0, 1]$. D29 was retained for protocol continuity and minimal intervention among 54 corrected-eligible candidates, not because it produced a preferred ITS winner or a favourable method comparison. The rejected families and the full candidate history are supplementary.

3.3 Oracle, Global Weights and the Decision Operator

Benefit-oriented criterion values are passed through $q_j(g) = \log(1 + 2g)/\log 3$. The noise-free oracle utility is

$$U_{as}^* = \frac{\sum_{j=1}^{10} \alpha_j q_j(g_{asj}) + \lambda \sum_{(j,k) \in \mathcal{E}} 0.20 q_j(g_{asj}) q_k(g_{ask})}{1 + \lambda}. \quad (4)$$

This is an internally specified reference, not causal truth and not social welfare. The prediction target adds controlled noise to U^* . Because Eq. (4) contains only main effects and pairwise interactions, the frozen interventional Shapley functional admits a closed-form *oracle attribution*. Global oracle weights normalize mean absolute WEIGHT-set attributions; learned SHAP applies the same aggregation to interventional TreeSHAP values from a fixed XGBoost model [5,7].

Six structured global-weight methods enter the gate: oracle attribution, SHAP, fixed-model context-block permutation importance (PI), non-negative Ridge+, CRITIC and entropy. Equal, RandomWeights and MajorityWinner serve as diagnostic references. Every weighting method produces exactly one context-invariant vector per world, which is the point: the design deliberately asks what survives that compression. Estimation uses disjoint FIT and WEIGHT partitions, and the 200 external TEST contexts are never used for fitting or weight estimation. At the Pilot-B size $N = 250$, each world holds 200 FIT, 50 WEIGHT and 200 TEST contexts.

The downstream operator is benefit-oriented weighted MOORA [19]:

$$r_{asj} = \frac{g_{asj}}{\sqrt{\sum_{b=1}^6 g_{bsj}^2 + 10^{-12}}}, \quad S_{as} = \sum_{j=1}^{10} w_j r_{asj}. \quad (5)$$

Table 2. The five prospectively frozen Pilot-B stop conditions. Any one of them halts the study; passage requires all five to be false.

Stop	Condition
Coverage	A required structured output or RandomWeights coverage is missing.
Random equivalence	No method clears median $\Delta\tau_b = 0.05$ or $\Delta\text{mean-regret} = 0.01$ versus RandomWeights.
Modal-winner tie	No method clears median $\Delta\text{Top-1} = 0.02$ or $\Delta\text{mean-regret} = 0.01$ versus MajorityWinner.
Oracle dominance	Pooled oracle modal share is at least 0.90.
Weight insensitivity	Random-weight choices or performance widths meet the frozen invariance criteria.

Alternatives are ranked by descending S_{as} . All score representations use an absolute 10^{-12} tie tolerance, anchor-based average ranks without tie chaining, and ascending alternative ID for deterministic Top-1 selection.

3.4 Decision Fidelity and the Prospective Adequacy Gate

Agreement with the oracle is measured three ways: full-ranking agreement by Kendall’s τ_b , deterministic Top-1 accuracy, and normalized oracle regret

$$R_s^{(q)} = \frac{U_{a_s, s}^* - U_{\hat{a}_s^{(q)}, s}^*}{\max_a U_{as}^* - \min_a U_{as}^* + 10^{-12}}. \quad (6)$$

Pilot B was frozen before execution at $(N, c, \rho, \lambda) = (250, 0.30, 0.4, 0.5)$ over development worlds 21001–21005. It uses 200 common Dirichlet(1, . . . , 1) RandomWeights draws per world and one MajorityWinner per world.

MajorityWinner is an oracle-informed diagnostic reference. It is the context-free oracle-modal alternative estimated on WEIGHT and evaluated on TEST—the constant alternative maximizing empirical oracle Top-1 frequency on WEIGHT. It is not claimed to minimize regret or to constitute an operational transportation policy, and it is neither stakeholder-derived nor normative. It is included precisely because it is strong and cheap: a global-weight pipeline should add decision value beyond a single constant choice before its additional complexity is considered justified.

A practical method advantage must be strictly positive in at least four of the five worlds *and* meet the relevant cross-world median margin. Table 2 lists the five independently sufficient stop conditions. The overall rule is an OR: the benchmark passes only if every stop is false. The margins are design-specific adequacy tolerances, not universal equivalence bounds.

The protocol, the evaluator and the protected-seed firewall were committed before the single authorized execution. A 30-world, 135-cell primary factorial and its associated robustness analyses were conditional on Pilot-B passage. They were

Table 3. Frozen Pilot-B outcome. The overall hard stop is the OR of the five flags; one flag fired.

Gate	Audited result	Stop
Coverage	Six structured methods eligible without fallback	No
Random equivalence	Four of six methods separated from RandomWeights	No
Modal-winner tie	Zero of six methods escaped MajorityWinner	Yes
Oracle dominance	Pooled modal share $0.526 < 0.90$	No
Weight insensitivity	Invariant fraction $0.004 < 0.90$; qualified worlds $0 < 4$	No
Overall	At least one prospectively frozen stop fired	Yes

not run. This is version-controlled prospective specification rather than public preregistration; the complete provenance appears in the supplement.

4 Results

4.1 Structural Validity Did Not Guarantee Decision Diversity

The historical generator of Eq. (1) collapsed numerically exactly where the algebra predicted. Across all 35 world–criterion combinations the maximum active-pair log-ratio variation was 7.2224×10^{-16} and the maximum vector-normalized profile variation was 1.8036×10^{-9} : machine noise, not signal.

The corrected D29 kernel cleared the frozen structural and reachability requirements, and it did produce multiple oracle orderings and multiple oracle winners in every world—between 17 and 78 distinct complete orderings and between two and five distinct winners. Yet its oracle decision geometry stayed markedly uneven. The modal oracle winner’s share ranged from 0.519 to 0.983 across the five worlds, with a median of 0.809. In world 21001, the directly comparable case, the correction raised the number of distinct oracle orderings from 19 to 23 while also raising modal share from 0.962 to 0.983.

Removing exact multiplicative separability was therefore necessary for this benchmark but not sufficient. In three of five worlds a single alternative was already oracle-optimal in over 80% of contexts, which sets a high bar for any pipeline that must beat a constant choice.

4.2 The Pilot-B Result

All six structured methods were defined in every world, every required Kendall summary was available, and no Equal fallback was used. Table 3 reports the independently reconstructed gate decision. Four of six methods separated from RandomWeights, so the random-equivalence stop was false. The pooled oracle modal share was 0.526 (A_2 in 526 of 1000 TEST contexts), so no single alternative dominated the pooled worlds, and RandomWeights decisions were not invariant. Exactly one stop fired: the modal-winner effective tie.

Table 4. Cross-world median Pilot-B contrasts, with the count of strictly positive worlds in parentheses (P, out of five). MW denotes MajorityWinner, RW RandomWeights. The last column reports the frozen RandomWeights separation verdict.

Method	MW	MW	RW	RW	RW sep.
	$\Delta\text{Top-1}$ (P)	Δregret (P)	$\Delta\tau_b$ (P)	Δregret (P)	
Oracle attribution	-0.010 (1)	-0.002119 (1)	0.209 (5)	0.023940 (4)	Yes
SHAP	0.000 (2)	0.000000 (2)	0.288 (5)	0.026060 (4)	Yes
Permutation importance	-0.120 (1)	-0.004425 (1)	0.194 (5)	0.010658 (4)	Yes
Ridge+	-0.010 (1)	-0.002119 (1)	0.237 (5)	0.011492 (4)	Yes
CRITIC	-0.580 (0)	-0.124414 (0)	0.030 (3)	-0.098354 (0)	No
Entropy	-0.580 (0)	-0.156641 (0)	0.035 (3)	-0.002807 (1)	No

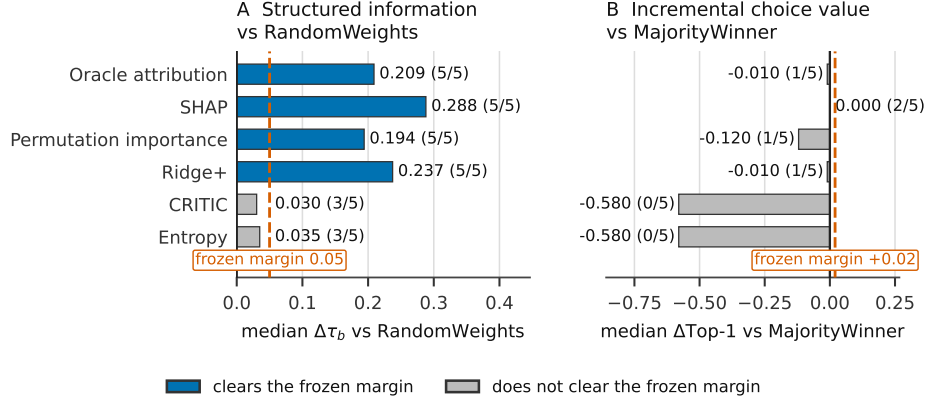


Fig. 2. Information learned is not decision value added. Left: every structured method except CRITIC and entropy clears the frozen 0.05 Kendall margin against RandomWeights. Right: no method reaches the frozen +0.02 Top-1 margin against the constant MajorityWinner reference, and SHAP's advantage there is exactly zero. Both panels are drawn directly from the frozen Pilot-B result on a common untruncated scale.

Table 4 and Fig. 2 localize the failure; positive values favour the structured method. Oracle attribution, SHAP, PI and Ridge+ each cleared a RandomWeights criterion with the required cross-world consistency. SHAP showed the largest median Kendall improvement, 0.288, positive in all five worlds, together with a median regret improvement of 0.026060, positive in four. Against MajorityWinner, by contrast, SHAP's median Top-1 and median regret advantages were both exactly zero, with only two positive worlds. No method met either modal-winner margin with the required four-of-five consistency.

The compact statement of the whole result is this. *Four structured methods, including SHAP, cleared the prospectively specified RandomWeights adequacy criterion, but no structured method cleared the prospectively specified MajorityWinner adequacy criterion.*

The exact oracle-attribution reference failed the same gate, so the stop cannot be attributed to XGBoost fitting error or TreeSHAP approximation alone.

Table 5. Post-closure descriptive diagnostic over the seven observed fixed vectors. Counts give worlds with a single selected alternative, and worlds with exact context-level agreement with MajorityWinner. Descriptive only; non-confirmatory.

Method	Constant-selection worlds	All-context MW-match worlds
Oracle attribution	1/5	1/5
SHAP	2/5	2/5
Permutation importance	0/5	0/5
Ridge+	1/5	1/5
CRITIC	2/5	1/5
Entropy	3/5	2/5
Equal	5/5	4/5

4.3 Post-Closure Descriptive Selection Maps

The Pilot-B artefact retained aggregate metrics but not method-level context selections. After permanent closure, a separate protocol was frozen before any selection was reconstructed. It reused the archived fixed weights and TEST identities, generated no target, fitted or queried no model, re-estimated no weight, and could not alter the gate; an independent audit verified all $5 \times 7 \times 200 = 7000$ selections. This is a post-closure descriptive diagnostic and has no confirmatory role.

Equal-weight MOORA selected a single alternative in every context of every world and matched MajorityWinner exactly in four of five worlds. Structured-method constancy was heterogeneous, from zero worlds for PI to three for entropy (Table 5). The simple hypothesis that every fixed-weight MOORA vector must be context invariant is therefore not supported for the seven observed vectors. World 21003 also shows why Top-1 and full-ranking fidelity must be reported separately: Equal, CRITIC and entropy all selected A_6 in all 200 contexts, giving identical Top-1 accuracy 0.190 and mean regret 0.316758, while their mean τ_b values still differed (0.5493, 0.5427, 0.5113) because the lower ranks differed. These observations suggest locally coarse decision regions for particular fixed vectors. They do not characterize unobserved weight space, do not establish a causal mechanism for the Pilot-B stop, and do not change its outcome.

5 Discussion

5.1 What the Study Establishes

RQ1 and RQ2 are addressed by the prospectively structured benchmark-development and Pilot-B evidence. The separately frozen post-closure diagnostic addresses DQ1 descriptively and has no confirmatory role.

For RQ1, an algebraic and numerical audit showed that contextual variation can vanish entirely under the normalization used by the decision operator, and that a diagnostic must be applied at that point to detect it. For RQ2, correcting

exact separability produced multiple oracle orderings and winners in every world, but it did not establish sufficient incremental decision value: no structured global-weight pipeline escaped MajorityWinner under the frozen cross-world rule.

The clearest reading is not that SHAP failed. Its separation from RandomWeights shows that its weights carried reproducible ranking information, and that entire median advantage then disappeared against a single constant choice: at this reference condition the bottleneck lay in converting one global weight vector into context-specific decision value through MOORA. Because closed-form oracle attribution reached the same outcome, an explanation resting solely on imperfect learned attribution is ruled out. No single causal mechanism is isolated, however: generator geometry, global compression, reference construction and MOORA geometry remain coupled in the executed design.

For applied OR and transport research the lesson is compact. A complex data-driven pipeline should be compared not only against random or uniform parameters but also against a meaningful simple decision policy: beating random weights shows that information was learned, not that it usefully changes the operational choice.

5.2 Interpretation Boundary and Limitations

The central boundary statement is this. *Pilot-B failure denotes failure of the prespecified adequacy requirement for the intended global-weight–MOORA comparison; it does not imply that the generator contains no discriminative decision information.* Four independent observations support the distinction: RandomWeights decisions were not invariant; four structured methods separated from RandomWeights; SHAP held a large and consistent Kendall advantage; and the observed fixed vectors displayed heterogeneous selection constancy rather than uniform collapse. Equally, the result does not show that SHAP universally fails, that global weights never matter, that MOORA is generally inadequate, that the D29 kernel is defective, that one ITS alternative is best, or that MajorityWinner should be deployed.

Ten limitations bound the claim.

1. Pilot B uses only five worlds, so cross-world inferential precision is low.
2. Those five are reused development worlds, not fresh ones.
3. Only one reference condition was evaluated, $(N, c, \rho, \lambda) = (250, 0.30, 0.4, 0.5)$.
4. *Prospective freezing prevents post-result modification of the gate, but it does not make the five reused development worlds independent validation data.*
The result is a protocol-level adequacy screen; no population-level inference about method performance is claimed.
5. MajorityWinner headroom varies strongly across worlds, from a modal share of 0.519 to 0.983, so the difficulty of the comparison is itself heterogeneous.
6. MajorityWinner is Top-1-derived, not regret-optimal. The regret arm uses the same prospectively frozen Top-1-derived constant reference rather than a separately optimized regret-minimizing constant policy. A future benchmark could prospectively define metric-matched constant references.

7. The generator, oracle and capability mask are synthetic assumptions, not empirical transport effects, causal truth, stakeholder preferences or social welfare.
8. Every global-weight method intentionally compresses context-specific information, and the methods differ in semantics: oracle attribution and SHAP target one interventional functional; Ridge+ and PI use supervised association; CRITIC and entropy summarize pooled decision-matrix properties. PI can additionally extrapolate when dependence is broken [11].
9. The post-closure diagnostic adds no independent worlds. A paired context bootstrap was prospectively excluded from that layer and was not executed; any resampling study would require its own protocol and could not reopen the gate.
10. The paper makes no empirical ITS deployment recommendation.

The negative result stays scientifically useful because it was not tuned away. The generator correction, the gate thresholds, the evaluator and the seed firewall all preceded the authorized result. Once the modal-winner stop fired, the 30-world primary factorial and the reserve worlds remained barred. Any future design must begin as a separately versioned study that discloses this closed result rather than retroactively amending it.

6 Conclusion

This study asked whether a synthetic ITS benchmark could support a global XAI-MCDM method comparison, and it asked before running the comparison at scale. Structural auditing found and corrected a normalization-induced collapse in the original generator, but that correction, while necessary, did not by itself establish downstream decision discrimination. At the frozen development reference, four structured methods—SHAP among them—cleared the RandomWeights adequacy criterion, yet none cleared the MajorityWinner criterion, and closed-form oracle attribution failed the same gate.

The defensible conclusion is narrow and actionable: at this reference condition the evaluated global-weight-compression-MOORA pipelines did not establish enough incremental decision value to justify the protected primary factorial, and the gate therefore performed its intended function. More broadly, synthetic XAI-MCDM studies should validate not only prediction and attribution, but also whether the benchmark can discriminate decisions beyond meaningful simple references. In XAI-assisted decision support, recovering structured importance is not enough; the recovered information must demonstrably change decisions in a useful way.

Code and Data Availability

The reproducibility package contains the synthetic benchmark generator, the frozen configuration and protocol files, the deterministic seed definitions, the

archived development-audit and Pilot-B result artefacts, the frozen global-weight vectors, the post-closure selection-map outputs, the independent audit and evaluator scripts, the software-environment specification, content-integrity manifests, and the read-only scripts that regenerate every table and figure reported here. Raw benchmark worlds are regenerated deterministically from the frozen seeds and configurations rather than stored as large duplicated datasets. Primary and reserve worlds were never accessed and no primary result exists. During double-blind review, identifying repository metadata is withheld; an anonymous archive can be supplied to editors and reviewers on request. The complete version-controlled repository and its immutable provenance record will be released publicly after review.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-Assisted Technologies

Generative AI tools were used for language editing, manuscript organization, consistency checking and LaTeX assistance. All scientific design decisions, code executions, numerical results, interpretations and citations were independently checked by the authors, who take full responsibility for the manuscript.

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